| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | | (ii) | | moles $MgCl_2 = \frac{0.384}{95} = 4.04 \times 10^{-3}$ mol<br><br>more **$Mg^{2+}$ ions** in second sample / magnesium chloride sample therefore it has more hardness (1)<br>accept more moles of magnesium chloride than magnesium sulfate (in the same mass because $M_r$ of magnesium chloride is smaller)<br><br>ecf possible from part (d) (i) | | | 2 | 2 | 1 | |
| | | | | **Question 10 total** | **2** | **2** | **8** | **12** | **3** | **6** |